

frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- symbolic computing
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- solving linear systems

3 Signals and Plants

- convolutions of functions
- plants defined by ordinary differential equations

MCS 472 Lecture 32
Industrial Math & Computation
Jan Verschelde, 3 April 2026

frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- symbolic computing
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- solving linear systems

3 Signals and Plants

- convolutions of functions
- plants defined by ordinary differential equations

the frequency domain

Definition (the Laplace transform)

Given an integrable function $f(t)$ in time t ,

the *Laplace transform* of $f(t)$ is $\mathcal{L}\{f\} = \int_0^{\infty} f(t)e^{-st}dt = F(s)$.

The Laplace transform takes a signal from the time domain, in t , to the *frequency domain*, using s as the symbol in the transform.

the transform of a linear combination

The Laplace transform of $f(t)$ is $\mathcal{L}\{f\} = \int_0^\infty f(t)e^{-st}dt = F(s)$.

The Laplace transform is linear.

- Let $f(t)$ and $g(t)$ be two integrable functions.
- Let a and b be two coefficients, independent of t .

Then

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s),$$

where

- $F(s)$ is the Laplace transform of $f(t)$, and
- $G(s)$ is the Laplace transform of $g(t)$.

delay and exponentials

The Laplace transform of $f(t)$ is $\mathcal{L}\{f\} = \int_0^\infty f(t)e^{-st}dt = F(s)$.

For some constant coefficient a , we have:

- $f(t - a)$ has Laplace transform $e^{-as}F(s)$
- $e^{-at}f(t)$ has Laplace transform $F(s + a)$
- $f(at)$ has Laplace transform $F(s/a)/a$

Exercise 1: Verify the above properties.

what is s ?

The Laplace transform of $f(t)$ is $\mathcal{L}\{f\} = \int_0^{\infty} f(t)e^{-st}dt = F(s)$.

Consider the Laplace transform of $f(t) = 1$:

$$\begin{aligned}\int_0^{\infty} 1e^{-st}dt &= \int_0^{\infty} \frac{-1}{s}e^{(-st)}d(-st) \\ &= -\frac{e^{-st}}{s}\Bigg|_0^{\infty} = 0 - \left(-\frac{1}{s}\right) = \frac{1}{s}.\end{aligned}$$

This leads to

$$s \int_0^{\infty} 1e^{-st}dt = 1 \quad \text{or} \quad s\mathcal{L}\{1\} = 1.$$

Multiplying with s can be called a *generalized differentiation*.

the Laplace transform of a derivative

The Laplace transform of $f(t)$ is $\mathcal{L}\{f\} = \int_0^\infty f(t)e^{-st}dt = F(s)$.

Proposition (the Laplace transform of the derivative)

$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = sF(s) - f(0), \quad F(s) = \mathcal{L}\{f\}$$

Apply integration by parts: $\int u dv = uv - \int v du$.

$$\begin{aligned}\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} &= \int_0^\infty e^{-st} \frac{d}{dt}f(t) = e^{-st}f(t) \Big|_0^\infty - \int_0^\infty f(t) \frac{d}{dt}e^{-st} \\ &= 0 - f(0) - \int_0^\infty f(t)(-s)e^{-st}dt \\ &= -f(0) + sF(s)\end{aligned}$$

higher order derivatives

The Laplace transform of $f(t)$ is $\mathcal{L}\{f\} = \int_0^\infty f(t)e^{-st}dt = F(s)$.

Proposition (the Laplace transform of the derivative)

$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = sF(s) - f(0), \quad F(s) = \mathcal{L}\{f\}$$

Corollary (the Laplace transform of the n -th derivative)

For $n > 0$:

$$\mathcal{L}\left\{\frac{d^n}{dt^n}f(t)\right\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0),$$

where $F(s) = \mathcal{L}\{f\}$.

frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- **symbolic computing**
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- solving linear systems

3 Signals and Plants

- convolutions of functions
- plants defined by ordinary differential equations

the Laplace transform in SymPy.jl

```
@syms s, t
one = sympy.laplace_transform(1, t, s)

returns (1/s, 0, True).

sympy.inverse_laplace_transform(one[1], s, t)

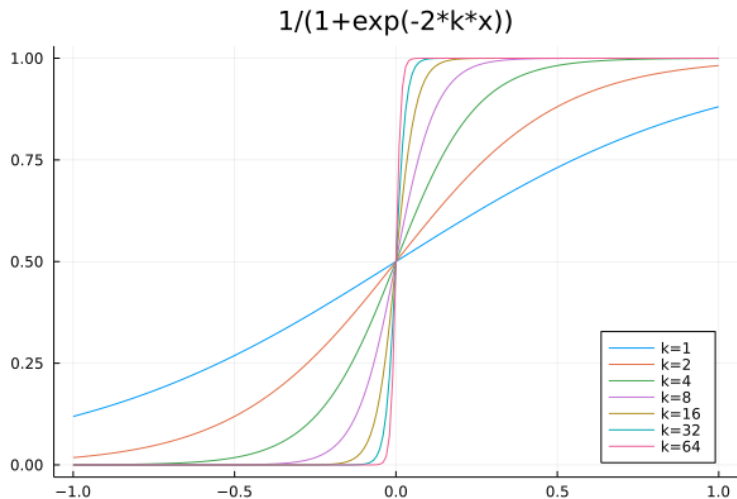
returns  $\theta(t)$ .
```

Definition (the Heaviside function)

The Heaviside function is the piecewise function

$$\theta(t) = \begin{cases} 1 & \text{if } t > 0, \\ 1/2 & \text{if } t = 0, \\ 0 & \text{if } t < 0. \end{cases}$$

the Heaviside function as $\lim_{k \rightarrow \infty} (1 + e^{-2kx})^{-1}$



frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- symbolic computing
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- solving linear systems

3 Signals and Plants

- convolutions of functions
- plants defined by ordinary differential equations

the inverse Laplace transform

The Laplace transform of $f(t)$ is $\mathcal{L}\{f\} = \int_0^{\infty} f(t)e^{-st} dt = F(s)$.

Definition (the inverse Laplace transform)

The *inverse Laplace transform* of $F(s)$ is f with the property

$$\mathcal{L}\{f\}(s) = F(s).$$

the Fourier transforms

The Laplace transform of $f(t)$ is $\mathcal{L}\{f\} = \int_0^{\infty} f(t)e^{-st} dt = F(s)$.

Observe the similarity with continuous Fourier transforms:

Definition (the Fourier transform)

The *Fourier transform* of f is $\mathcal{F}\{f\} = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt = F(i\omega)$.

Definition (the inverse Fourier transform)

The *inverse Fourier transform* of F is $\mathcal{F}^{-1}\{F\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(i\omega)e^{i\omega t} d\omega$.

frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- symbolic computing
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- solving linear systems

3 Signals and Plants

- convolutions of functions
- plants defined by ordinary differential equations

solving ordinary differential equations

To solve an Ordinary Differential Equation (ODE), do:

- 1 Transform the ODE into the frequency domain.
- 2 Solve the algebraic equation in s .
- 3 Transform the solution into the time domain.

an example

$$\frac{d^2}{dt^2}y(t) + 3\frac{d}{dt}y(t) + 2y(t) = 1, \quad y(0) = 0, \quad y'(0) = 0.$$

- 1 Denoting $\mathcal{L}\{y\} = Y(s)$, the transformed equation is

$$s^2 Y(s) + 3sY(s) + 2Y(s) = \frac{1}{s}.$$

- 2 We solve the algebraic equation: $s(s+1)(s+2)Y(s) = 1$.

$$Y(s) = \frac{1}{s(s+1)(s+2)} = \frac{1/2}{s} - \frac{1}{s+1} + \frac{1/2}{s+2}$$

- 3 Transformed back into the time domain, the solution is

$$y(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}.$$

damped harmonic motion

A model of a suspension mechanism, using the equation for damped harmonic motion, written in engineering notation:

$$\frac{d^2}{dt^2}y(t) + 2\zeta\omega \left(\frac{d}{dt}y(t) \right) + \omega^2 y(t) = 0,$$

with initial conditions:

$$y(0) = 1 \quad \text{and} \quad y'(0) = 0.$$

Exercise 2:

Apply the 3-step operational calculus method to compute the solution of this model.

solving a system

Consider

$$\begin{cases} x' &= -2x - 9y \\ y' &= x - 2y \end{cases} \quad x(0) = -1, \quad y(0) = 1.$$

- 1 Denoting $X(s) = \mathcal{L}\{x\}$ and $Y(s) = \mathcal{L}\{y\}$, the transformed system is

$$\begin{cases} sX + 1 &= -2X - 9Y \\ sY - 1 &= X - 2Y \end{cases} \quad \text{or} \quad \begin{cases} (s+2)X + 9Y &= -1 \\ -X + (s+2)Y &= 1 \end{cases}$$

- 2 Use `SymPy` to solve the system.

- 2 Use SymPy to solve the system.

$$\begin{cases} (s+2)X + 9Y = -1 \\ -X + (s+2)Y = 1 \end{cases}$$

```
@syms X, Y
eq1 = (s+2)*X + 9*Y + 1
eq2 = -X + (s+2)*Y - 1
sol = solve([eq1, eq2], [X, Y])
```

has output

```
Dict{Any, Any} with 2 entries:
  Y => (s + 1)/((s + 2)**2 + 9)
  X => -(s + 11)/(s**2 + 4*s + 13)
```

transform into the time domain

Dict{Any, Any} with 2 entries:

$$Y \Rightarrow (s + 1) / ((s + 2) ** 2 + 9)$$

$$X \Rightarrow -(s + 11) / (s ** 2 + 4 * s + 13)$$

③ Transform the solution into the time domain.

```
xt = sympy.inverse_laplace_transform(sol[X], s, t)
```

returns $-e^{-2t}(3 \sin(3t) + \cos(3t))\theta(t)$ as $x(t)$.

```
yt = sympy.inverse_laplace_transform(sol[Y], s, t)
```

returns $-e^{-2t}(\sin(3t)/3 + \cos(3t))\theta(t)$ as $y(t)$.

Note: for $t > 0$, we have $\theta(t) = 1$ and can drop $\theta(t)$.

frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- symbolic computing
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- **solving linear systems**

3 Signals and Plants

- convolutions of functions
- plants defined by ordinary differential equations

solving linear systems

Let A be an n -by- n matrix and consider the system

$$\frac{d}{dt}\mathbf{z}(t) = A\mathbf{z}, \quad \mathbf{z}(0) = \mathbf{z}_0.$$

- 1 Denoting $\mathcal{L}\{\mathbf{z}\} = \mathbf{Z}(s)$, the transformed equation is

$$s\mathbf{Z}(s) - \mathbf{z}_0 = A\mathbf{Z}(s).$$

- 2 The solution in the frequency domain is

$$\mathbf{Z}(s) = (sI - A)^{-1}\mathbf{z}_0,$$

where I is the identity matrix.

the characteristic values

- ② If A has k distinct eigenvalues, then

$$\det(sI - A) = \prod_{i=1}^k (s - \lambda_k)^{m_k},$$

where m_k is the multiplicity of the k th eigenvalue λ_k .

The matrix partial fraction expansion has then the form:

$$(sI - A)^{-1} = \sum_{k=1}^r \sum_{j=1}^{m_k} \frac{v_{j,k}}{(s - \lambda_k)^j}.$$

- ③ Use the partial fraction expansion to transfer into the time domain.

solving a system with eigenvalues

Exercise 3: Consider again

$$\begin{cases} x' &= -2x - 9y \\ y' &= x - 2y \end{cases} \quad x(0) = -1, \quad y(0) = 1.$$

- 1 Write the system in matrix format.
- 2 Compute the eigenvalues of the matrix.
- 3 Compare the eigenvalues with the previously computed solution, previously computed with the aid of `SymPy`.

frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- symbolic computing
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- solving linear systems

3 Signals and Plants

- **convolutions of functions**
- plants defined by ordinary differential equations

convolutions of functions

Definition (convolutions of functions)

Let f and g be functions of t . The **convolution** $f \star g$ of f and g is

$$f \star g = \int_0^t f(\tau)g(t - \tau)d\tau.$$

Recall the discrete convolution $y_k = \sum_{i=0}^k h_i u_{k-i}$, where

- y_k is the k th component of the output signal, $y = h \star u$,
- h is the response of the unit impulse, and
- u is the input signal.

applied to continuous signals

For continuous input $u(t)$, unit impulse response $h(t)$,

$$y(t) = h(t) \star u(t)$$

is the continuous output of the filter.

In the frequency domain:

$$Y(s) = H(s)U(s)$$

where the Laplace transforms are

$$U(s) = \mathcal{L}\{u\}, \quad H(s) = \mathcal{L}\{h\}, \quad \text{and} \quad Y(s) = \mathcal{L}\{y\}.$$

plant

Definition (plant)

A *plant* is a linear, time invariant, causal map, which maps signals into signals.

Examples: filters, mechanisms, circuits.

By the linearity, time invariance, and causality, the plant is determined by its response to the unit impulse.

frequency domains, signals and plants

1 The Frequency Domain

- Laplace transforms
- symbolic computing
- Laplace and Fourier

2 Operational Calculus

- solving ordinary differential equations
- solving linear systems

3 Signals and Plants

- convolutions of functions
- plants defined by ordinary differential equations

plants defined by ordinary differential equations

Consider the plant defined by

$$\frac{d}{dt}y(t) + ay(t) = u(t),$$

where $u(t)$ is the input and $y(t)$ the output.

$$\begin{aligned}\frac{d}{dt} \left(e^{at} y(t) \right) &= ae^{at} y(t) + e^{at} \frac{d}{dt} y(t) \\ &= e^{at} \left(ay(t) + \frac{d}{dt} y(t) \right) = e^{at} u(t)\end{aligned}$$

$$\frac{d}{dt} \left(e^{at} y(t) \right) = e^{at} u(t) \quad \Rightarrow \quad y(t) = e^{-at} \left(\int_0^t e^{a\tau} u(\tau) d\tau + y(0) \right)$$

convolutions and the Laplace transform

$$\begin{aligned}y(t) &= e^{-at} \left(\int_0^t e^{a\tau} u(\tau) d\tau + y(0) \right) \\&= \int_0^t e^{-a(t-\tau)} u(\tau) d\tau + y(0)e^{-at}\end{aligned}$$

Set $y(0) = 0$. The output of the plant is then

$$y(t) = e^{-at} \star u(t).$$

In the frequency domain:

$$Y(s) = \left(\frac{1}{s+a} \right) U(s).$$

summary and bibliography

The Laplace transform turns differential equations into algebraic ones. We defined the convolution of continuous functions, made the connection between signals and ordinary differential equations.

We started Chapter 10 of our text book.

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

An introductory text book in control has many examples to engineering:

- Richard C. Dorf and Robert H. Bishop:
Modern Control Systems. Ninth Edition, Prentice Hall, 2001.